Week 6 Part 2 Final Assignment Sekou Ruddock

Dataset Recap

I have processed and profiled a historical emergency department dataset containing over 560,000 distinct patient encounters. For each encounter, I analyzed approximately 225 patient characteristics, ranging from basic demographics (age, gender) to specific vital signs (heart rate, oxygen saturation, temperature) to predict the patient's official Emergency Severity Index (ESI) on a scale of 1 (resuscitation) to 5 (non-urgent clinic care). The dataset also includes 200 one-hot encoded (OHE) features representing specific chief complaints. To improve data interpretability, we have grouped these into clinically distinct families, such as cardiovascular, respiratory, traumatic, and neurological presentations. This structure reveals potential confounders, as overlapping symptoms such as "shortness of breath" occurring in both cardiac and respiratory families can obscure the primary etiology of a patient’s condition. Furthermore, the high dimensionality of these OHE flags introduces complexity, as patients frequently present with multiple, co-occurring complaints that act as confounding variables when attempting to isolate specific clinical risks. These flags remain highly sparse, as many patients do not have a standard complaint checked off in the system. Overall, the patient mix skews heavily toward mid-acuity (ESI 3) cases, which accurately reflects a standard emergency department caseload.

Model Descriptions

Logistic Regression

Logistic Regression serves as our baseline model to assist with triage categorization. Rather than treating all patient data as equal, the model calculates the relative importance of each clinical sign such as specific vital signs or symptom flags and aggregates them into a final risk probability for the patient.

Essentially, the model converts these combined inputs into a percentage chance of a patient requiring urgent care. It then sorts patients into triage streams based on a calculated threshold. We standardized the clinical data before training to ensure a fair assessment; this prevents metrics with large numerical ranges (like heart rate) from inadvertently overpowering critical but numerically smaller signs (like age), allowing every clinical factor to contribute proportionally to the final decision.

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Decision Tree

A Decision Tree model functions like an automated, logical flow chart. It guides the triage process by asking a sequence of clear, binary questions before arriving at a recommended triage level. It provides a clear, traceable path for every decision therefore it does not act as a “black box” . Clinicians can review this logic step-by-step, allowing us to see exactly which factors influenced the recommendation and verify that the output aligns with our established ED protocols

Benchmark Table

Model	**Accuracy**	**Macro F1**	**Recall (ES1)**
Dummy (Baseline)	0.375	0.204	0.000
Logistic Regression	0.671	0.484	0.188
Decision Tree	0.556	0.216	0.000

Metric Justification

When evaluating the triage models, I prioritized Recall for ESI 1 above all other metrics. In an Emergency Department setting, the clinical consequences of a "False Negative"where the model fails to identify a patient who needs immediate resuscitation are potentially life-threatening.

While Accuracy and Macro F1 provide a high-level view of model performance, they can be misleading in clinical settings where certain classes (like ESI 1) are rare. A model could achieve high overall accuracy by correctly triaging the more common, mid-acuity cases while completely ignoring the rare, high-acuity ones. Therefore, we evaluate these models based on their ability to minimize "Missed Critical" cases (False Negatives), even if this results in more "False Alarms" (False Positives) for lower-acuity patients.

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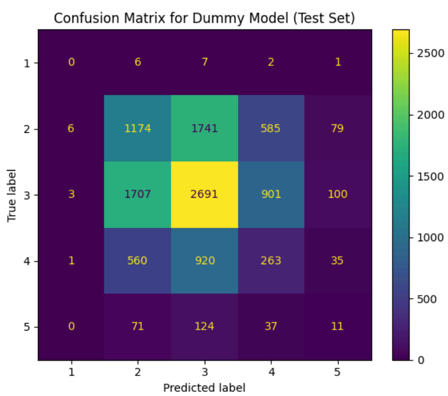
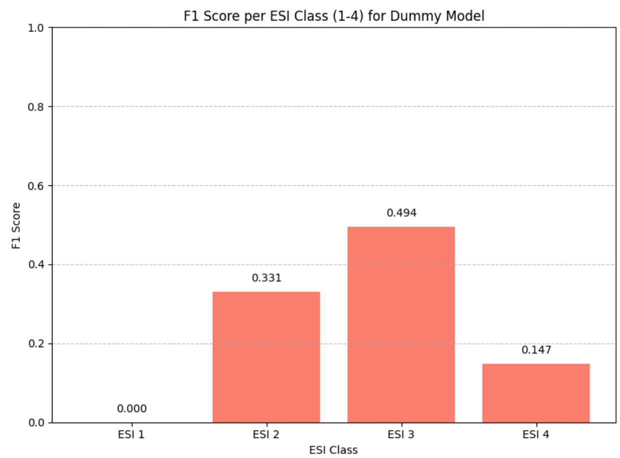
Failure Modes

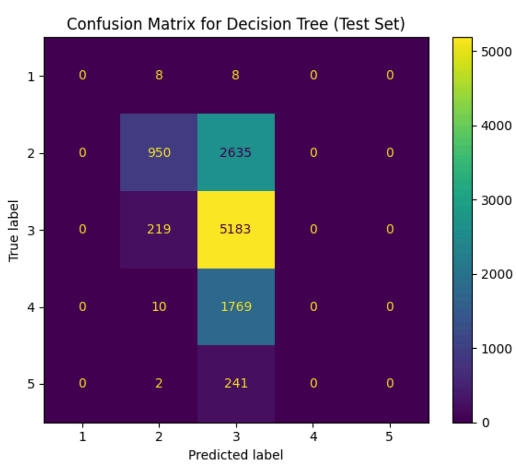
The most critical failure across all models is their inability to safely triage ESI 1 (resuscitation) patients. This is largely because ESI 1 cases are rare in our dataset, meaning the models have seen very few examples of these critical presentations during training.

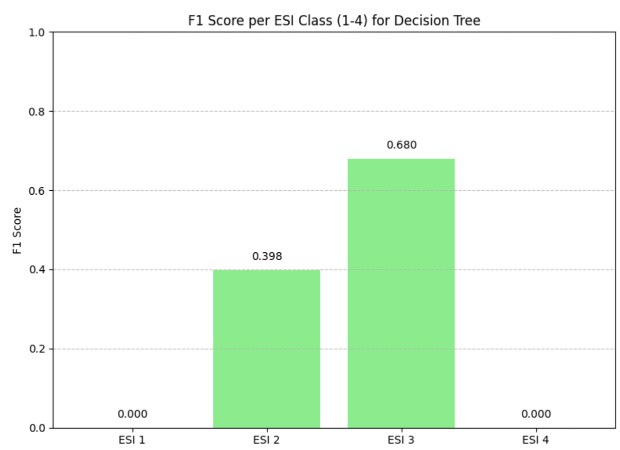
Dummy Model (Baseline):





Decision Tree: This model is effectively blind to the most critical patients, achieving 0.000 Recall for ESI 1. As shown in the confusion matrix and Histogram, it predicts zero ESI 1 cases correctly. Reliance on this model would lead to catastrophic delays in care for patients requiring resuscitation.





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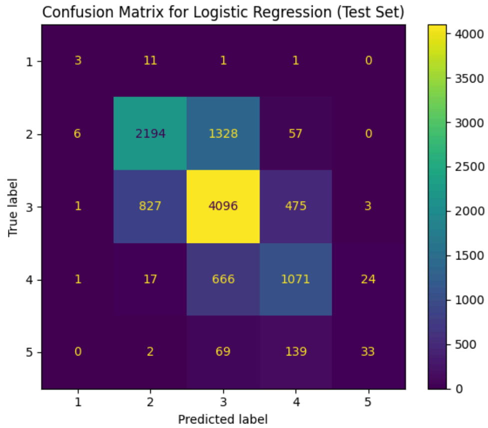
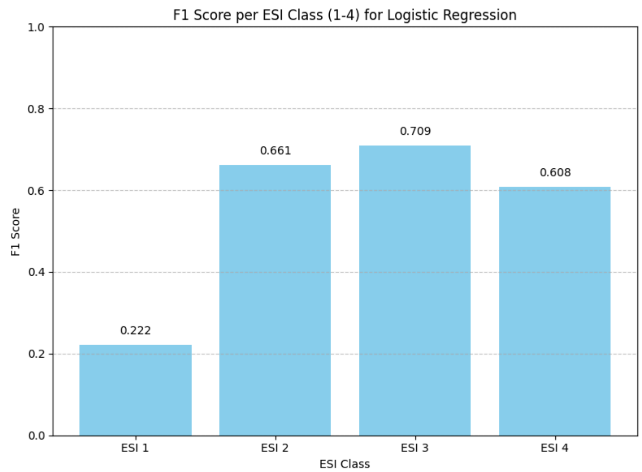
Logistic Regression: While this model outperforms the others, it only achieves 18.8% Recall for ESI 1. This means that for every 10 resuscitation cases, the model would miss 8 of them, leaving the majority of our sickest patients unidentified and at risk of delayed care.





Clinical Risk: Both models demonstrate high confusion between ESI 2 and ESI 3 cases, often down-triageing patients. The models are currently not safe for clinical deployment; they prioritize "correctly clearing" low-acuity cases while failing to catch the patients who need life-saving intervention.

Github Repo - https://github.com/Colosolizer/carisurg-portfolio